

## Anti-Forensics & Log Clearing

### What anti-forensics is

Actions deliberately taken to hinder forensic examination: deleting artifacts, altering timestamps, filling free space, tampering logs. Legit use: your own data cleanup and IR planning. Remember: in authorized engagements, cleanup timing and scope are dictated by RoE; elsewhere it is a crime.

### Classic techniques

| Technique                     | Detail                                                                                 |
|-------------------------------|----------------------------------------------------------------------------------------|
| Timestamping                  | Set file times to plausible past (PowerShell touch; Manipulation on Windows)           |
| Artifact removal              | Delete registry keys, prefetch, AppData, USNs (USN Journal edit)                       |
| Log tampering                 | Change/truncate \$LogFile, event logs, web logs                                        |
| Antivirus/persistence mirrors | Disable tools, stop services                                                           |
| Free-space wiping             | Overwrite slack/unallocated (sdelete-style, <code>dd if=/dev/zero</code> misses slack) |
| Memory untargeting            | Avoid disk writes; live-memory-only                                                    |

### Why it fails (the forensic truth)

**Timelines rebuild:** multiple timestamps (created/modified/accessed), \$MFT, USN journal, prefetch → inconsistencies expose the stomp

**Backups & logs elsewhere:** SIEM/hub-side logs survive a bricked workstation

**Minifilters & memory:** deleted ≠ unrecoverable; in-memory archeology (Volatility) shows what ran

**Cached artifacts:** Thumbs.db, pagefile/hiberfil, Volume Shadow Copies

**The attack itself rewrites history:** the attacker's own evasion steps create NEW artifacts; forensic analysts read the *differences*

## The analyst's reconstruct toolset

```
# timeline from artifacts (sleuth kit)
fls -r -m / mount.dd > body
mactime -b body > timeline.txt

# memory view
vol3 -f mem.dump windows.psscan # see processes deleted from lists
```

And the discipline: analyze many sources, hypothesize, verify.

Anti-forensics is not 'make it invisible'; it's 'make it complicated'.  
The complicated part is the story forensic specialists reconstruct anyway.

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[← Anonymization: VPN, Tor, Pro...](#)

Module complete ✓